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# Beyond EEG: MAESTRO, a Multimodal Dataset for Decoding Auditory Attention from Eye, Brain, and Body

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## Abstract

1 We release MAESTRO, a multimodal auditory-attention-decoding (AAD) corpus  
2 in which 16 normal-hearing adults completed 100 cued-attention trials each while  
3 listening to four concurrent spatialised talkers at  $\pm 22.5^\circ$  and  $\pm 67.5^\circ$  plus back-  
4 hemisphere noise at  $\pm 135^\circ$ . Synchronously recorded streams: 32-channel 500 Hz  
5 EEG, binocular gaze and pupillometry, head-mounted scene video, head IMU, the  
6 per-trial stimulus audio for each device, and a comprehension-question response.  
7 Existing AAD corpora deliver EEG and stimulus envelopes only; MAESTRO is  
8 the first AAD release that ships gaze, IMU, and head-mounted scene video on the  
9 same trials, enabling the “is this really an auditory-cortex signal or just overt orien-  
10 tation?” question to be answered quantitatively rather than rhetorically. Around  
11 the data we define three task framings (attended hemisphere, attended eccentricity,  
12 4-class attended source) and an envelope-reconstruction track, all packaged in a ver-  
13 sioned evaluation protocol with four splits (within-subject, leave-one-subject-out,  
14 motion-residualised, high-quality tier), six decision windows for continuous AAD,  
15 and three mandatory baselines. Headline findings on this protocol: (i) classical  
16 envelope-reconstruction TRF/CCA is at the chance floor of the paradigm (0.523  
17 pooled hemisphere with the best of four backbones); (ii) a 368-feature classical-  
18 spectral EEG classifier reaches 0.716 [0.641, 0.792], while a gaze-only baseline  
19 reaches 0.770 [0.728, 0.811]; (iii) a strict trial-level motion-residualisation that  
20 orthogonalises EEG features against gaze+IMU+video collapses spectral-EEG  
21 hemisphere AAD to 0.521 [0.488, 0.552], exposing a confound-controlled floor  
22 that any future “purely neural” decoder must beat; (iv) at the trial level, gaze-  
23 based AAD tracks comprehension success ( $r=+0.54$ ) whereas envelope-tracking  
24 decoders *anti-track* it (Wilcoxon  $p=0.034$  on the split  $\delta+\theta$  TRF), dissociating  
25 effortful acoustic tracking from delayed comprehension. The dataset, loaders,  
26 60+ reproducible baseline results, and Croissant metadata record are hosted at an  
27 anonymous URL during review.

## 1 Introduction

29 Auditory attention decoding (AAD) is the problem of inferring, from non-invasive recordings of  
30 a listener, which of several concurrent talkers they are attending to. Public datasets have driven  
31 progress by releasing EEG plus the attended and unattended audio: KUL AAD Das et al. [2019],  
32 DTU AAD Fuglsang et al. [2018], and recent MEG-AAD releases cover most published decoders.  
33 These datasets capture speech-tracking EEG but *not* the behavioural channels that generate the  
34 recurring reviewer concern that spatial AAD is entangled with overt orientation: pupil dilation  
35 follows arousal, gaze trajectories lateralise toward the attended source Zion Golumbic et al. [2013],

and subtle head orientation tracks effortful listening Hendrikse et al. [2019]. Without synchronous gaze and head-motion streams it is not possible to ask, on the same trials that support the AAD claim, how much of the classifier is neural and how much is overt behaviour.

**What this dataset adds.** For every trial we have (i) 32-ch 500 Hz EEG; (ii) binocular gaze direction, pupillometry, and scene-projected gaze; (iii) 100 Hz head-mounted IMU; (iv) 25 Hz egocentric scene video; (v) the stereo audio played through each of three speaker devices; and (vi) a comprehension question answered after every trial. No previously released AAD corpus bundles all five non-audio streams.

**What we do with it.** We treat the corpus not as a bag of recordings but as the substrate for a *contract*: the set of tasks, splits, windows, and baselines a paper must report on if it claims an AAD result on this data. Around that contract we ship reproducible baselines on three task framings and a strict motion-residualisation control that gives a confound-controlled floor against which any “purely neural” decoder must compete. We use the multimodality to settle—quantitatively, on the same trials—two long-running questions: *is the EEG decoder eye-tracking in disguise?* and *do AAD decoders track comprehension or effort?*

**Contributions.** (i) A 16-subject, 1600-trial AAD corpus with five simultaneously recorded modalities, including two (head-mounted scene video and IMU) never previously released with AAD (Sec. 3). (ii) A versioned evaluation protocol with three task framings, four splits, six decision windows, and three mandatory baselines, packaged as a one-call Python harness (Sec. 4). (iii) Reproducible baseline benchmarks across envelope-reconstruction TRF/CCA, a 368-feature classical-spectral EEG classifier, gaze/IMU/scene-video unimodal decoders, early/late/calibrated multimodal fusion, SNR- and window-stratified sweeps, learning curves, and leave-one-subject-out transfer (Sec. 5). (iv) Diagnostic controls that exploit the multimodality: an aggressive cleaning ablation (Sec. 6.1), a symmetric cross-modal predictability map between EEG, gaze, IMU, video, and audio (Sec. 6.2), a strict trial-level motion-residualisation that yields a confound-controlled floor (Sec. 6.3), and a trial-level behaviour- $\times$ -decoder analysis that dissociates effortful acoustic tracking from comprehension (Sec. 6.4). (v) Croissant metadata, a Hugging Face / OpenNeuro mirror, and per-experiment SLURM scripts that reproduce every number in the paper from the raw recordings.

**Scope.** This is a dataset and benchmarks paper, not a new modelling paper. The shipped baselines are deliberately standard—linear TRF, CCA,  $\ell_2$ -logistic regression on classical features, LightGBM, and reliability-weighted late fusion—so the numbers in Sec. 5 represent a faithful floor, not a tuned ceiling, for future work. Deep-learning decoders are implemented as drop-in slots behind a RUN\_DEEP flag but are not run in this submission (Sec. 9).

## 2 Related AAD datasets

Public AAD corpora (Table 1) cover  $\leq 44$  subjects and ship EEG-plus-audio only; Fuglsang et al. [2020] adds in-ear EEG and listening-effort labels. The community has raised gaze, pupil dilation, and subtle head orientation as confounds on every AAD generation Zion Golumbic et al. [2013], Hendrikse et al. [2019], Geirnaert et al. [2021, 2024], and recent MEG-AAD releases still cover only a single non-neural channel. No previously released AAD corpus bundles synchronised EEG, binocular eye-tracking *and* pupillometry, head-mounted IMU, *and* egocentric scene video on the same trials. This is the gap MAESTRO addresses.

Table 1: AAD dataset modality coverage. EEG: high-density scalp. Gaze: 2-D/3-D eye-tracking. IMU: inertial. Video: head-mounted scene camera. Q/A: per-trial comprehension question.

| Corpus                         | Subj.     | Trials                    | Spkrs          | EEG | Gaze | IMU | Video | Audio | Q/A |
|--------------------------------|-----------|---------------------------|----------------|-----|------|-----|-------|-------|-----|
| KUL AAD Das et al. [2019]      | 16        | 8 trials of 6 min         | 2              | ✓   | —    | —   | —     | ✓     | —   |
| DTU AAD Fuglsang et al. [2018] | 18        | 60 trials of 50 s         | 2              | ✓   | —    | —   | —     | ✓     | —   |
| <b>MAESTRO (ours)</b>          | <b>16</b> | <b>100 trials of 30 s</b> | <b>4+noise</b> | ✓   | ✓    | ✓   | ✓     | ✓     | ✓   |

### 77 3 Dataset

78 16 adult native English speakers (self-reported normal hearing) completed one session each. After five  
79 familiarisation trials the participant was cued, on each of 100 evaluation trials, to attend to one of four  
80 spatial talkers while three other competing streams played simultaneously for 30 s. After every trial  
81 the participant answered a multiple-choice comprehension question about the cued stream; per-trial  
82 correctness is released as an attention-compliance label (mean compliance 0.82, range [0.61, 0.92];  
83 full psychometric in Appendix A.5). Demographic questionnaires were collected only in a voluntary  
84 opt-in follow-up (5/16 subjects, ages 18–30, mixed gender); to preserve participant anonymity during  
85 review we release only the aggregate range.

86 **Spatial geometry.** Six speakers form a horizontal ring at a radius of 4 ft. Device 1 carries the two  
87 left-hemisphere speech speakers at azimuth  $-67.5^\circ$  (Device-1 L) and  $-22.5^\circ$  (Device-1 R); Device 2  
88 carries the right-hemisphere pair at  $+22.5^\circ$  (Device-2 L) and  $+67.5^\circ$  (Device-2 R); Device 3 plays  
89 back-hemisphere multi-talker noise at  $\pm 135^\circ$ . Each device plays one stereo file with its two channels  
90 driving its left and right speakers at different powers; per-channel power and attended-speaker SNR  
91 are reported in `audio_stimuli_data/trials.csv`. SNR varies from 3 dB to 18 dB (median  
92 12 dB, IQR 10 dB to 14 dB). The attended-speaker label is one of  $\{1, 2, 3, 4\}$ .

93 **Modalities recorded per trial.** 32-channel 500 Hz EEG (ANT Neuro *eego mylab*, 10–20 montage);  
94 binocular gaze direction, pupillometry, and scene-projected gaze at 50 Hz (Tobii Pro Glasses 3);  
95 100 Hz head-mounted IMU; 25 Hz,  $1920 \times 1080$  egocentric scene video; the stereo audio played  
96 through each of three devices. Hardware models, channel montage, sampling rates, storage layout,  
97 and the cross-modal temporal-alignment procedure are detailed in Appendices A.1, A.2, and A.3.

### 98 4 Tasks and evaluation protocol

99 A multimodal release is only useful to the community if different papers’ numbers on it are comparable.  
100 We commit to protocol v1.0 (versioned; future revisions preserve the v1.0 split so no number  
101 ever becomes orphaned). Every published accuracy on this dataset must report the (task, split,  
102 window, subject-subset) tuple; our harness `scripts/evaluation_protocol.py` enforces this by  
103 construction.

#### 104 4.1 Tasks

105 We define three classification framings over the four-speaker label, plus a continuous envelope-  
106 reconstruction track. **All three classification framings are required** in any paper reporting AAD on  
107 this dataset.

- 108 • **T1 attended hemisphere** (chance 0.5):  $\{1, 2\}$  (left,  $\leq -22.5^\circ$ ) vs.  $\{3, 4\}$  (right,  $\geq +22.5^\circ$ ).  
109 The classical binary AAD framing. Future spatial-AAD methods that report only this number  
110 are interpretable but cannot be compared to ones that exploit eccentricity or source identity.
- 111 • **T2 attended eccentricity** (chance 0.5):  $\{2, 3\}$  (inner,  $\pm 22.5^\circ$ ) vs.  $\{1, 4\}$  (outer,  $\pm 67.5^\circ$ ).  
112 Tests whether the decoder picks up eccentricity rather than hemisphere; gaze-based decoders  
113 show a much smaller eccentricity gradient than hemisphere gradient, so T1–T2 is a useful  
114 diagnostic of where a decoder’s signal lives.
- 115 • **T3 4-class attended source** (chance 0.25). The full label. The hardest task and the one  
116 that most strongly distinguishes information that is genuinely about source identity from  
117 information that is merely about side or distance.
- 118 • **T4 envelope reconstruction** (continuous; metric  $\rho_{\text{att}} - \rho_{\text{una}}$  and binary attendance accuracy  
119 at decision windows  $\{1, 2, 4, 8, 16, 30\}$  s). The classical TRF/CCA target. We report it  
120 because it is the lingua franca of AAD methods, and we report its bit-rate (information  
121 transfer rate) at every window so that real-time methods can be benchmarked against the  
122 same throughput axis.

## 4.2 Splits

- **S1 within-subject 5-fold stratified CV** (seed 0). The headline number. Reported as the unweighted mean across the 16 per-subject accuracies with bootstrap 95 % CI (10000 resamples over subjects).
- **S2 leave-one-subject-out (LOSO)**. Generalisation across subjects. Quantifies the per-subject calibration premium any deployable method must close.
- **S3 motion-residualised within-subject** (Sec. 6.3). The confound-controlled floor: a method claiming purely neural gains must move *this* number, not (or not only) S1. The shipped harness performs the residualisation inside every CV fold so no train/test motion-information leaks.
- **S4 high-quality-tier within-subject**: subjects with comprehension  $\geq 0.80$  and Tobii gaze-validity  $\geq 0.70$  (pre-registered criteria, not post-hoc accuracy). Reports an optimistic ceiling while making the inclusion rule transparent; 8/16 subjects pass at v1.0.

## 4.3 Windows, metrics, and mandatory baselines

The canonical decision window for T1–T3 is the full 30-s trial; continuous-AAD submissions must additionally report  $\{1, 2, 4, 8, 16, 30\}$  s accuracy on T4 and the corresponding ITR. Primary metric is accuracy with bootstrap 95 % CI; secondary metrics are per-subject scatter and ITR. A methods paper comparing to MAESTRO must report the three following baselines (reproducible by one command each, Appendix A.16):

- **B1 Gaze-only logistic regression (S1, T1)**. Hemisphere target 0.770. The first number any new EEG decoder must clear before claiming a multimodal contribution.
- **B2 EEG-spectral 368-D logistic regression (S1, T1)**. Hemisphere target 0.716. The non-deep ceiling.
- **B3 EEG-spectral on motion-residualised features (S3, T1)**. Hemisphere target 0.521. *Any EEG-method gain over B3 is the paper’s headline.*

Multimodal methods must additionally report a *marginal-gain* number: S1 accuracy minus the better of (a) gaze-only and (b) EEG-spectral-only. Without this, a multimodal paper cannot tell whether its gain comes from fusion or from one dominant modality; our fusion baselines (Sec. 5.4) make the point that pooled fusion can hide per-subject gain, and the protocol forces future papers to surface that.

## 5 Benchmarks

### 5.1 Envelope reconstruction (TRF/CCA): the floor of this paradigm

We fit ridge backward TRFs and CCA over EEG lags  $[-200, +500]$  ms against the 28-band gamma-tone envelope of the attended audio, with four backbones (broadband Hilbert envelope 1–9 Hz; split  $\delta + \theta$ ; mel-28; CCA on mel-28). Ridge  $\alpha$  is selected by inner 5-fold CV on  $10^{-1} \dots 10^6$  scored by  $\rho_{\text{att}} - \rho_{\text{una}}$ .

Table 2: Linear TRF / CCA envelope-attendance accuracy at the 30-s window, S1 (within-subject 5-fold CV, chance 0.5). *Wins* = number of subjects on which this backbone is best.

| Backbone                                     | Pooled       | 95 % CI        | Wins subj |
|----------------------------------------------|--------------|----------------|-----------|
| CCA mel-28 (inner CCA + logistic on $\rho$ ) | <b>0.523</b> | [0.502, 0.544] | 9/16      |
| Mel-28 ridge TRF                             | 0.511        | [0.486, 0.538] | 2/16      |
| Broadband ridge TRF                          | 0.507        | [0.476, 0.537] | 4/16      |
| Split $\delta + \theta$ ridge TRF            | 0.492        | [0.459, 0.524] | 1/16      |
| LOSO CCA mel-28 (S2)                         | 0.539        | [0.516, 0.562] | –         |
| Oracle adaptive (inner-CV pick per subject)  | 0.555        | –              | –         |

Linear envelope reconstruction sits at the chance floor of the paradigm. Short windows ( $\leq 4$  s) stay at chance; accuracy climbs only to 0.539 at 30 s (window sweep in Appendix A.9). The shallow

gradient is a designed property: each device plays a single stereo file whose two channels drive its two speakers at different powers, so the envelope shape is near-identical between the two sides. Envelope reconstruction therefore cannot pick between the two speakers of the same device and collapses to chance on T3 (mean 0.255, chance 0.25). **Implication:** classical envelope-AAD decoders are a reporting requirement (T4) on this dataset, not a competitive headline. The paradigm decouples spatial source identity from envelope content—a feature for studying spatial attention in the multimodal regime, but a limitation for porting two-talker AAD decoders unchanged.

## 5.2 Classical spectral EEG classifier

We compute a 368-dimensional feature vector per trial on 1–40 Hz preprocessed EEG at 500 Hz: log Welch band powers in  $\delta/\theta/\alpha/\beta/\gamma$  per channel (160), within-channel band ratios (96), four region-wise alpha-lateralisation indices, Hjorth mobility/complexity per channel (64), Shannon spectral entropy per channel (32), region-wise  $1/f$  aperiodic slope (4), alpha-band MSC and wPLI within-left, within-right, and between-hemisphere (6), and global field power mean and std (2).  $\ell_2$ -logistic regression with 5-fold stratified CV per subject.

Table 3: Classical EEG spectral-feature classifier, S1 (within-subject), pooled over 16 subjects.

| Task           | L2-LogReg                   | LightGBM             | Gaze-only (ref.)     |
|----------------|-----------------------------|----------------------|----------------------|
| T1 hemisphere  | <b>0.716</b> [0.641, 0.792] | 0.655 [0.589, 0.724] | 0.770 [0.728, 0.811] |
| T2 inner-outer | <b>0.649</b> [0.592, 0.712] | 0.597 [0.536, 0.667] | 0.683 [0.597, 0.766] |
| T3 4-class     | <b>0.448</b> [0.358, 0.544] | 0.410 [0.332, 0.497] | 0.547 [0.429, 0.664] |

The 368-feature linear classifier reaches 0.716 pooled hemisphere AAD—+21.8 pp over the best linear-TRF backbone. LightGBM under-performs, characteristic of  $p=368$  vs.  $n\approx 100$  per subject; regularised linear is the right default at this dataset size. Nine of 16 subjects pass 0.70 hemisphere; five pass 0.88. **Why this matters for AAD methodology:** the dominant published AAD decoder family (envelope reconstruction) is the wrong question to ask on this paradigm; a 368-D linear classifier on classical spectral and connectivity features is at +21.8 pp over it. Future deep-EEG methods on this data should benchmark against B2 (0.716), not against the envelope-TRF floor.

## 5.3 Unimodal behavioural baselines: gaze, IMU, scene video

A 23-dimensional per-trial gaze+IMU summary (scene-projected mean/std, per-eye azimuth/elevation, pupil diameter, sample validity, saccade rate and median amplitude, 3-D gaze centroid, head rotation magnitude) feeds  $\ell_2$ -logistic regression. **Gaze alone reaches 0.770 [0.728, 0.811] pooled hemisphere accuracy;** 11/16 subjects exceed 0.70. Eccentricity (T2) is weaker (0.683): gaze is strongly lateralising but only weakly eccentricity-encoding. T3 gaze-only reaches 0.547 (+30 pp over chance), and gaze LOSO on T3 transfers to 0.475 [0.405, 0.548], the strongest cross-subject number in the release.

A 35-feature IMU baseline (per-axis accel/gyro stats plus dominant motion frequencies and spectral-band ratios) pools at 0.607 [0.541, 0.670] on T1; a 15-feature scene-video baseline (motion energy, brightness, dense optical-flow stats on a  $160\times 90$  downsample) pools at 0.617 [0.558, 0.675]. The two are nearly identical—as expected, a head-mounted scene camera in a quiet room is largely a derivative of head motion. On T2 both rise to  $\approx 0.66$ , beating the EEG spectral classifier (0.649) on that task: the eccentricity cue is mostly carried by head orientation. **This is why MAESTRO ships head-motion modalities:** without IMU and head-mounted video, a 0.61–0.66 confound floor on these tasks would be invisible.

## 5.4 Multimodal fusion

*Early:* concatenate  $[\text{EEG}_{368}, \text{gaze}_{23}]$  and feed LightGBM ( $n=300$ ,  $\text{lr} = 0.05$ ). *Late (mean):* average per-modality logistic posteriors. *Late (calibrated):* weight posteriors by each modality’s inner-CV accuracy above chance. *Stacked:* a second-stage logistic regression on concatenated posteriors.

Pooled fusion does *not* exceed gaze-only. **But per-subject fusion does help when both modalities are strong:** S7 hemisphere  $0.89 \rightarrow 0.97$ ; S15  $0.92 \rightarrow 1.000$ ; S2  $0.91 \rightarrow 0.95$ . The calibrated late-fusion scheme never drops below either unimodal baseline because its weights collapse to the stronger

Table 4: Pooled fusion accuracy vs. best unimodal baselines (S1, chance shown next to task). Early/Late/Stacked numbers are EEG CCA-mel + gaze; Late-cal is calibrated late-fusion of the spectral-LogReg posteriors with gaze, weighted by inner-CV accuracy above chance.

| Task (chance)        | EEG spec. | Gaze         | Early (LGBM) | Late (mean) | Stacked | Late-cal |
|----------------------|-----------|--------------|--------------|-------------|---------|----------|
| T1 hemisphere (0.5)  | 0.716     | <b>0.770</b> | 0.764        | 0.768       | 0.758   | 0.745    |
| T2 inner-outer (0.5) | 0.650     | <b>0.683</b> | 0.660        | 0.672       | 0.665   | 0.687    |
| T3 4-class (0.25)    | 0.448     | <b>0.547</b> | 0.513        | 0.525       | 0.510   | 0.509    |

modality when the weaker one is near chance. This is the headline finding for which a multimodal  
release is the right unit of analysis: pooled averaging hides per-subject gain. We therefore report  
per-subject scatter alongside every pooled number (Fig. 1).

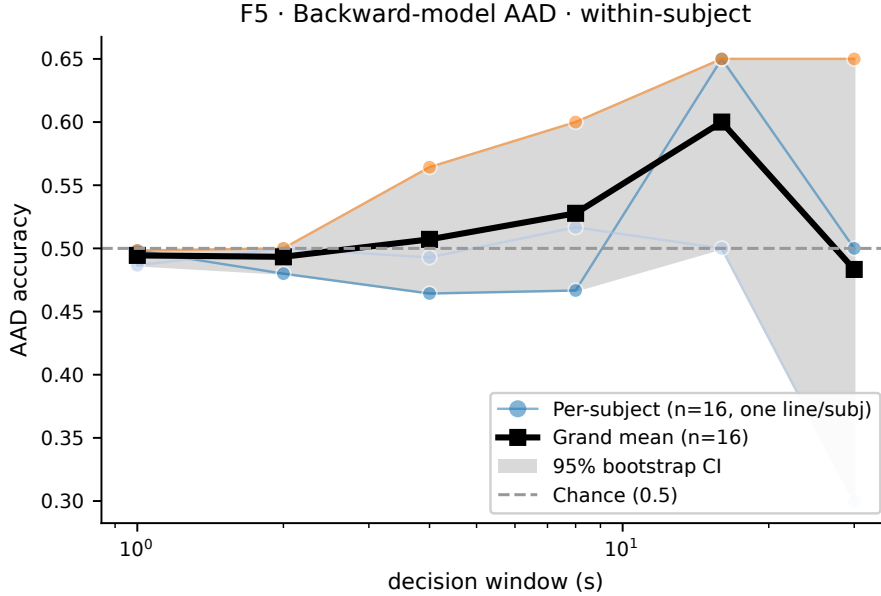


Figure 1: Per-subject AAD accuracy as a function of decision-window length for the backward-TRF CCA mel-28 envelope decoder (S1, T4). Each thin coloured line is one of the 16 subjects (legend bundles them). The thick black line with squares is the grand mean across subjects with the shaded band marking its 95 % bootstrap CI; the dashed line marks chance (0.5). Per-subject scatter widens with window length, justifying per-subject reporting alongside any pooled number.

## 5.5 Generalisation sweeps: SNR, window, learning curve, LOSO

Four stress tests are first-class results and required entries for the protocol (Sec. 4); summary numbers below, full tables in Appendices A.8–A.14.

- **SNR stratification.** Envelope-CCA hemisphere accuracy rises monotonically from 0.464 at  $\leq 6$  dB to 0.525 at  $\geq 15$  dB (+6.1 pp). The spectral classifier peaks in the middle bins (0.670 at 7–10 dB)—weakest at low SNR (poor neural SNR despite high listening effort) and at high SNR (where the task is nearly trivial and listeners may lapse). Gaze accuracy is nearly flat across SNR, consistent with gaze indexing the spatial-attention decision rather than acoustic SNR per se.
- **Window sweep.** Envelope CCA stays at chance for  $\leq 4$  s and reaches 0.539 at 30 s; the spectral classifier is defined only at the 30-s trial window. Continuous-AAD methods must report all six windows.
- **Learning curves.** Gaze-only saturates by  $n_{\text{train}}=40$ ; the spectral classifier is still gaining between  $n=60$  and  $n=80$ , implying additional accuracy is recoverable from longer sessions. Pooled early fusion stays below gaze-only at every  $n$ , confirming the Sec. 5.4 finding.

223 • **LOSO (S2)**. Envelope CCA transfers at 0.539; gaze 23-D at 0.715; the spectral classifier  
 224 at 0.615 (drops 10.1 pp vs. S1, quantifying the per-subject calibration premium); concate-  
 225 nated EEG+gaze at 0.696. Gaze dominates cross-subject because between-subject EEG  
 226 topography variation is much larger than between-subject gaze-behaviour variation.

## 227 6 Diagnostics: what is the EEG decoder actually decoding?

228 The point of a multimodal release is that every confound a reviewer can name has a synchronous  
 229 ground-truth recording on the same trials. We use that to answer four questions any AAD methods  
 230 paper using MAESTRO will face.

### 231 6.1 Cleaning ablation: does the signal survive aggressive nuisance removal?

232 We re-run the 368-feature classifier under five EEG preparations: *raw* (Sec. 5.2); *ica\_eog* (ICA with  
 233 automatic EOG-component removal using Fp1/Fpz/Fp2 as proxies); *regress\_gaze* (channel-wise  
 234 ridge regression of 13 Tobii-gaze features out of each EEG channel); *regress\_gaze\_imu* (same plus 8  
 235 IMU features); *drop\_eog\_emg* (exclude Fp1/Fpz/Fp2 as EOG sources and T7/T8/F7/F8/FC5/FC6 as  
 236 EMG sources, leaving 23 channels).

Table 5: Spectral classifier (S1) under five EEG-cleaning conditions. Largest pooled drop on hemisphere is  $-3.9$  pp.

| Condition        | T1 hemisphere               | T2 inner-outer       | T3 4-class           |
|------------------|-----------------------------|----------------------|----------------------|
| raw              | <b>0.716</b> [0.643, 0.789] | 0.649 [0.593, 0.712] | 0.448 [0.359, 0.543] |
| ica_eog          | 0.681 [0.614, 0.752]        | 0.649 [0.594, 0.713] | 0.440 [0.359, 0.529] |
| regress_gaze     | 0.695 [0.619, 0.770]        | 0.644 [0.587, 0.711] | 0.449 [0.359, 0.548] |
| regress_gaze_imu | 0.709 [0.637, 0.783]        | 0.644 [0.581, 0.711] | 0.457 [0.368, 0.553] |
| drop_eog_emg     | 0.677 [0.601, 0.747]        | 0.641 [0.580, 0.708] | 0.437 [0.347, 0.535] |

237 Every cleaning step retains  $\geq 0.67$  hemisphere AAD; channel-wise gaze+IMU regression costs only  
 238  $-0.7$  pp ( $0.716 \rightarrow 0.709$ ); the largest drop ( $-3.9$  pp) comes from removing 9 EOG-/EMG-prone  
 239 channels entirely. The remaining 23 central/parietal/occipital channels are exactly where auditory  
 240 cortex projects (Cz, Pz, CP1/2, C3/4, P3/4, POz, O1/Oz/O2). **So the signal is not in the EOG-prone**  
 241 **channels and does not vanish under standard nuisance removal at the time-series level.** This is  
 242 the answer to the simplest version of “is this just eye-tracking?”—no.

### 243 6.2 Cross-modal predictability: where does EEG variance live?

244 We turn the question around: take the same 368-D EEG features and ask how well they linearly  
 245 predict each individual feature of gaze, IMU, video, and audio (per-subject 5-fold ridge regression,  
 246  $\lambda=10$ , held-out Pearson  $r$ , pooled across 16 subjects). Top features per modality are summarised in  
 247 Table 6; the full 93-feature table is in Appendix A.6.

Table 6: Top EEG-decodable feature per modality (held-out Pearson  $r$ , pooled over 16 subjects, 5-fold ridge  $\lambda=10$ ).

| Modality (#feats) | Top feature               | Pooled $r$ | 95 % CI            |
|-------------------|---------------------------|------------|--------------------|
| Gaze (23)         | <b>sacc_rate</b>          | +0.549     | [+0.467, +0.630]   |
| IMU (35)          | <b>gyr_mag_mean</b>       | +0.559     | [+0.468, +0.647]   |
| Video (15)        | <b>motion_energy_mean</b> | +0.487     | [+0.403, +0.565]   |
| Audio (20)        | <b>una_env_der_pos</b>    | $-0.086$   | $[-0.139, -0.038]$ |

248 EEG spectral content is *strongly* linearly related to head-motion amplitude, gaze validity, pupil  
 249 diameter, and scene-motion energy ( $r \in [0.45, 0.56]$  for the top feature in each), and *essentially*  
 250 *unrelated* to per-trial audio summary statistics ( $|r| \leq 0.09$ ). This reads at first like a contradiction  
 251 with the cleaning ablation (Sec. 6.1)—if EEG and motion share  $r \approx 0.55$ , isn’t AAD just motion in  
 252 disguise? The two findings together resolve it cleanly: *a substantial fraction of EEG variance is*

253 *motion- or arousal-correlated and is genuinely predictable from external sensors, but the attended-*  
 254 *speaker-discriminating component lives in a smaller subspace that the cleaning ablation does not*  
 255 *touch.* The next subsection isolates that subspace explicitly. The directional flow features (which-way  
 256 did the camera pan) and the per-trial audio summaries are at zero predictability from EEG: EEG  
 257 encodes *motion amplitude*, not motion direction, and the AAD signal is not “EEG looks like the  
 258 spectrogram” (which would require time-resolved TRF analysis on aligned series, not a per-trial  
 259 summary).

### 260 6.3 Trial-level motion-residualisation: a confound-controlled floor

261 The strict control: force the trial-level EEG feature vector to be linearly independent of the trial-level  
 262 motion feature vector. Per CV fold, we fit  $M = \text{Ridge}(\alpha=10)$  on  $Z_{\text{train}} \rightarrow X_{\text{train}}$  where  $Z \in \mathbb{R}^{n \times 71}$   
 263 is the concatenated gaze (23-D) + IMU (35-D) + video (13-D) summary and  $X \in \mathbb{R}^{n \times 368}$  is the EEG  
 264 spectral feature matrix; we then replace  $X$  on both train and test with their residuals  $\tilde{X} = X - M(Z)$   
 265 and refit the AAD classifier. We compare to a *shuffle-motion* null in which  $Z$  is row-permuted across  
 266 trials—a control that removes a 71-D subspace with no trial-level binding to  $X$  and thus measures  
 267 the capacity cost of orthogonalisation per se.

Table 7: Motion-residualisation. Spectral classifier on raw vs. residualised EEG features. Wilcoxon  $p$  paired against *raw*,  $n=16$ . The lower CI of motion-residualised T1 is below chance.

| Task           | raw                         | shuffle-motion                       | motion-residualised                                 |
|----------------|-----------------------------|--------------------------------------|-----------------------------------------------------|
| T1 hemisphere  | <b>0.716</b> [0.641, 0.793] | 0.643 [0.585, 0.701] ( $p=10^{-3}$ ) | <b>0.521</b> [0.488, 0.552] ( $p=6 \cdot 10^{-4}$ ) |
| T2 inner-outer | 0.650 [0.592, 0.714]        | 0.610 [0.559, 0.665] ( $p=0.018$ )   | 0.510 [0.480, 0.536] ( $p=10^{-3}$ )                |
| T3 4-class     | 0.442 [0.357, 0.535]        | 0.392 [0.328, 0.462] ( $p=0.017$ )   | 0.270 [0.239, 0.296] ( $p=4 \cdot 10^{-3}$ )        |

268 Orthogonalising with a 71-D *shuffle* costs 7 pp on T1 (capacity cost of removing any 71-D subspace  
 269 from a 368-D feature space). Orthogonalising with the *real* motion features costs a further 12 pp—the  
 270 motion-attributable component. The residualised number (0.521, lower CI 0.488) sits at the chance  
 271 floor: **a substantial fraction of the discriminative information in the 368-D classifier lives in a**  
 272 **subspace that co-varies with head and gaze orientation.**

273 **How to interpret this.** This is neither an artefact discovery nor a refutation of the AAD signal.  
 274 Attending to a lateralised sound source produces both lateralised neural activity *and* lateralised subtle  
 275 head/gaze orientation; the two are aspects of the same attentional process and they share variance.  
 276 What this control delivers is a confound-controlled floor (0.521, chance 0.5) that future EEG-only  
 277 or multimodal methods can target: a paper that claims a “purely neural” AAD gain on this dataset  
 278 must move *this* number, not the raw S1 number alone. Reconciling with Sec. 6.1: time-domain  
 279 channel-wise regression (which costs  $-0.7$  pp) and trial-level feature-space residualisation (which  
 280 costs  $-19.5$  pp) answer different questions—the latter can remove signal that the former cannot reach  
 281 because the classifier and the residualiser operate in the same trial-summary basis. A continuous  
 282 sweep  $X_\alpha = X - \alpha M(Z)$  for  $\alpha \in [0, 1]$  shows a smooth, monotone, approximately linear cost curve  
 283 (Appendix A.11), consistent with genuine entanglement rather than a clean separable subspace.

### 284 6.4 Behaviour $\times$ decoder: comprehension vs. effort

285 For every decoder we computed  $\Delta_s = P(\text{decoder correct} \mid \text{behav. correct}) - P(\text{decoder correct} \mid$   
 286  $\text{behav. wrong})$  per subject, then a signed Wilcoxon test across 16 subjects.

Table 8: Within-subject behaviour- $\times$ -decoder paired test.  $\Delta_s$  is the per-subject difference in decoder accuracy between trials where the subject answered the comprehension question correctly vs. wrongly.

| Decoder                              | Task                | Median $\Delta_s$ | Sign (+/−) | Wilcoxon $p$ |
|--------------------------------------|---------------------|-------------------|------------|--------------|
| Backward-TRF split $\delta + \theta$ | envelope-attendance | −0.084            | 3 / 13     | <b>0.034</b> |
| Backward-TRF broadband               | envelope-attendance | −0.068            | 5 / 11     | 0.093        |
| CCA mel-28 (gaze-residualised)       | envelope-attendance | −0.087            | 5 / 11     | 0.117        |
| Gaze 23-D                            | T1 hemisphere       | +0.068            | 11 / 5     | 0.044        |
| Spectral 368-D                       | T1 hemisphere       | +0.037            | 9 / 7      | 0.180        |



287 Two opposite-sign findings sit beside each other: (i) gaze-based AAD is positively correlated with  
288 comprehension correctness ( $p=0.044$ ); (ii) envelope-tracking AAD is *negatively* correlated with  
289 comprehension correctness, reaching nominal  $p<0.05$  on the split  $\delta + \theta$  TRF. The 368-D spectral  
290 classifier sits in between (median  $\Delta_s = +0.037$ ,  $p=0.18$ ) and effectively dissociates from both signs.

291 **What this means for AAD methodology.** Gaze-based decoders mechanically track the  
292 comprehension-relevant orienting behaviour: subjects who comprehend more fixate the attended  
293 speaker more consistently. Envelope-tracking TRF decoders read out an acoustic cortico-speech  
294 alignment that the listener can sustain *without* successfully recalling the question—and may in  
295 fact *recruit more* on trials where comprehension is hard, consistent with effort/arousal accounts of  
296 low-frequency speech tracking. Two practical consequences for users of MAESTRO: (a) a paper  
297 should not claim that a TRF-based AAD decoder reflects “successful comprehension”—the trial-level  
298 data rule that out; (b) the same dataset can therefore be used to develop *listening-effort* probes (split  
299  $\delta + \theta$  TRFs as a candidate) that are statistically dissociable from comprehension-correlated gaze  
300 decoders.

## 301 7 Dataset access, licensing, and Croissant metadata

302 **Hosting.** The full corpus ( $\approx 650$  GB uncompressed, dominated by scene video) is hosted on a  
303 BIDS-formatted OpenNeuro mirror for EEG plus a Hugging Face Datasets record for the Tobii bundle  
304 and audio; both are linked from an anonymous landing page during review. A  $\approx 9$  GB *sample* (3  
305 subjects, first 10 trials each, all modalities) is mirrored on an anonymous URL for reviewer inspection.  
306 Code and loaders are at an anonymous GitHub URL with `environment.yml`, per-notebook run  
307 instructions, SLURM submission scripts, and the `evaluation_protocol` harness.

308 **License.** Audio stimuli: drawn from the public-domain LibriVox corpus (CC-0), redistributable  
309 as-is. EEG, Tobii bundle (gaze+IMU+scene video), and behavioural metadata: released under  
310 CC-BY-NC-4.0 to allow academic reuse while preserving the identifiability caveats of scene-camera  
311 footage (see Ethics).

312 **Croissant metadata.** We ship a `croissant.json` at the repository root including both the core  
313 `RecordSet` definitions for every trial artefact and the Responsible AI (RAI) fields required by the  
314 NeurIPS E&D track (`dataUseCases`, `dataPersonalSensitiveInformation`, `dataAnnotationProtocol`,  
315 `dataCollectionType`). The Croissant record validates against the current schema Akata et al. [2024].

316 **Reference pipeline.** Twelve Jupyter notebooks plus a shared `aad_utils` package: data audit,  
317 behavioural analysis, EEG quality, gaze, audio features, TRF/CCA decoding, gaze-only decoding,  
318 cross-modal predictability, fusion, publication figures, scene-video analysis, and advanced EEG  
319 neuroscience. Each notebook records the parquet artefacts that feed downstream. A `RUN_DEEP` flag  
320 (off by default) stubs out GPU-backed deep-model evaluations.

## 321 8 Ethics and broader impact

322 The study was approved by the Institutional Review Board (protocol number and institution disclosed  
323 in the camera-ready version). All participants gave written informed consent specifically covering  
324 the release of de-identified EEG, eye-tracking, IMU, and head-mounted scene video. Scene video  
325 is the only modality capable of re-identifying participants or bystanders: we release the head-  
326 mounted egocentric footage (no participant face), with faces of incidental persons in the lab blurred  
327 by an automated pipeline (MediaPipe face detector at threshold 0.3 followed by manual review).  
328 Participants were compensated at the minimum-wage rate in the study’s jurisdiction.

329 **Intended use.** The corpus is released for research on multimodal attention decoding, hearing-aid  
330 algorithms, and brain–computer interfaces. Foreseeable misuses include using per-trial behavioural  
331 data to infer trait-level attributes of participants, and applying AAD decoders in surveillance contexts  
332 to infer listening targets; the CC-BY-NC-4.0 licence and access-by-request gate for the raw scene  
333 video are our mitigations.

## 9 Limitations

- **Cohort size.** 16 subjects is adequate for within-subject AAD, matches the KUL corpus, and exceeds what is standard for new-paradigm AAD releases; it is however small for cross-subject (LOSO) generalisation studies and we report LOSO numbers as a calibration floor, not a deployment target.
- **Paradigm-specific.** The stereo-device design (one stereo file per device, two channels driving two speakers at different powers) makes within-device L/R discrimination near-impossible for envelope-reconstruction decoders and is what drives the chance-floor numbers in Sec. 5.1. This is a feature for studying spatial attention in the multimodal regime (it decouples source identity from envelope content) but a limitation for porting classical two-talker AAD decoders unchanged.
- **One subject (S1) has hardware-bad mastoids.** Spherical-spline interpolation recovers a valid 32-ch trace, but S1 remains an outlier on several metrics. S1 is included in all pooled numbers; the high-quality-tier split S4 excludes S1 via a pre-registered gaze-validity criterion, never by post-hoc accuracy.
- **No deep-model ceilings in this release.** All headline numbers are linear or low-capacity. Deep EEG decoders—EEGNet, a 1-D CNN stimulus reconstructor, a LaBraM/BIOT foundation-model probe, and a cross-modal attention module—are implemented in `scripts/deep_models.py` behind a `RUN_DEEP` flag; running them is blocked on GPU allocation and will be added in a v1.1 analysis-pack update without changing any dataset bits.
- **Shared-variance confound.** The motion-residualisation control (Sec. 6.3) shows that a substantial part of the spectral-classifier signal shares variance with head/gaze orientation. We interpret this as expected (shared cause: spatial attention drives both neural and orienting responses), but downstream work should prefer the motion-residualised floor when claiming purely neural gains.
- **Demographics partially opt-in.** Age and gender were collected only in a voluntary follow-up; we release only the aggregate range and do not stratify analyses by demographics.

## 10 Conclusion

MAESTRO is a 16-subject, 1600-trial AAD dataset with five simultaneously recorded modalities, including head-mounted gaze, IMU, and scene video—modalities conspicuously missing from the existing AAD public record. Around the data we ship a versioned evaluation protocol with three task framings, four splits, six windows, and three mandatory baselines, plus a Python harness that enforces the contract. The benchmarks expose four findings that only a multimodal release can produce: (i) classical envelope-reconstruction is at the chance floor of the paradigm and a 368-D linear spectral classifier is +21.8 pp above it; (ii) gaze alone outperforms either EEG decoder pooled, but per-subject calibrated late fusion never under-performs the stronger modality; (iii) strict trial-level motion-residualisation collapses the spectral-classifier signal to chance, defining a confound-controlled floor that any “purely neural” decoder must beat; (iv) at the trial level, gaze-based AAD tracks comprehension while envelope-tracking AAD anti-tracks it, dissociating effortful acoustic attention from delayed comprehension. The dataset gives the community the first AAD corpus on which the “auditory attention or overt orientation” question can be answered quantitatively on the same trials that produce the AAD number, and the first on which a new multimodal decoder can be scored against both a pooled accuracy target and a motion-orthogonal one.

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## 402 A Technical appendices and supplementary material

### 403 A.1 Recording hardware and sampling rates

- 404 • **EEG.** ANT Neuro *eego mylab* amplifier, 32-channel 10–20 montage (Fp1, Fpz, Fp2, F7, F3,  
 405 Fz, F4, F8, FC5, FC1, FC2, FC6, M1, T7, C3, Cz, C4, T8, M2, CP5, CP1, CP2, CP6, P7,  
 406 P3, Pz, P4, P8, POz, O1, Oz, O2) at 500 Hz with linked-mastoid online reference; ADC clip  
 407 at 0.0839 V.
- 408 • **Eye tracking, IMU, scene camera.** Tobii Pro Glasses 3 worn throughout the session.  
 409 The authoritative gaze stream is the per-eye 3-D gaze direction plus pupil diameter and  
 410 scene-projected gaze2d from `gazedata.gz` (50 Hz). IMU is `imudata.gz` at 100 Hz. Scene  
 411 camera is `scenevideo.mp4` at 25 Hz, 1920×1080. Experiment-data 2-D gaze files are *not*  
 412 used in the current pipeline.
- 413 • **Audio.** Three stereo devices with fixed spatialisation (Sec. 3); per-trial stimulus files in  
 414 `audio_stimuli_data/pairs/*.flac`. `audio_stimuli_data/trials.csv` lists per-  
 415 trial attended-speaker index, the three device files, attended-speaker SNR, and per-channel  
 416 powers.

### 417 A.2 Storage layout and trial indexing

418 Per-subject, two directory trees hold parallel copies of the same trials partitioned by  
 419 modality: `experiment_data/Subject N/` (EEG and audio playback timestamps) and  
 420 `Video Recordings/Subject N/` (Tobii bundle: gaze, IMU, scene video, event data). Fold-  
 421 ers 1–5 are training trials, 6–105 are main trials. Subjects 1–2 use literal numeric folder  
 422 names (1...105). Subjects 3–16 use ISO-8601 timestamps (YYYYMMDDTHHMMSSZ); the loader  
 423 parses the embedded time and sorts ascending to recover the natural recording order, verified  
 424 empirically against audio playback timestamps to  $r_{\Delta} \geq 0.99$  trial-to-trial for every subject  
 425 (`results/01_alignment_check.parquet`).

### 426 A.3 Cross-modal temporal alignment

427 All streams are clipped to the shared audio-playback window  $[t_0, t_1]$  recovered from each trial’s  
 428 `audio_timestamps.json`. EEG’s internal clock `eeg_ts` is mapped to wall-clock time via the  
 429 `first_sample_time` field in `eeg_time_data.p`. Tobii recording-relative timestamps are shifted to

align their first sample to  $t_0$ . Audio envelopes are computed from each device’s stimulus file and resampled to the target analysis rate (commonly 64 Hz to match the EEG low-pass). The alignment sanity check is shipped in `analysis/01_data_audit.ipynb`.

#### A.4 Data quality audit (EEG)

**Automatic bad-channel detection.** A channel is flagged bad if (i) its standard deviation is below  $10^{-9}$  V (flat), (ii) at least 10 % of samples reach the 0.0839 V ADC clip (saturation), or (iii) its AC variance has  $|z| > 6$  relative to the median-absolute-deviation of the remaining channels. Per-subject and per-channel bad-rates are in `results/bad_channels_manifest.parquet` (Fig. 2). The dominant failure mode is mastoid saturation: Subject 1 has 100 % flat M1 and 100 % saturated M2, and M2 saturation is common cohort-wide.

**Robust referencing.** Linked mastoids when both M1/M2 are clean; common-average over the good channels otherwise; flagged channels then spherical-spline interpolated. Every trial therefore has 32 valid channels entering downstream analyses regardless of raw electrode issues.

**Gaze tracking quality.** Tobii per-eye validity (the fraction of samples where the tracker locked a gaze vector) exceeds 90 % for 14/16 subjects and is reported per-trial in `results/07_gaze_features.parquet`. Trials with  $< 50$  % validity are flagged but not removed from the main release.

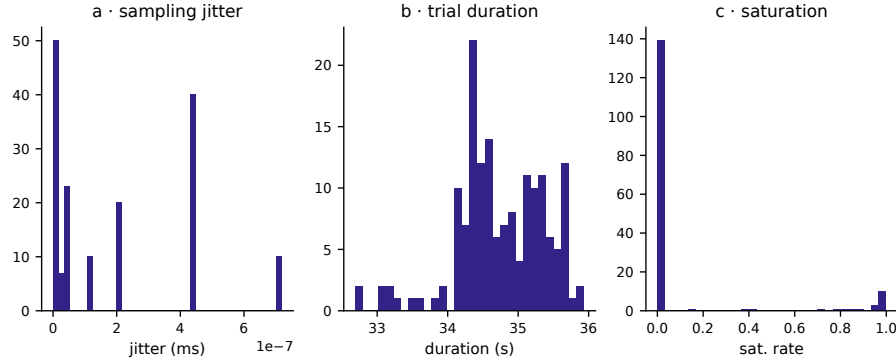


Figure 2: EEG quality. Left: bad-channel rate by channel (M1/M2 carry most failures). Centre: PSDs with well-formed  $1/f$  and alpha peak at Pz/POz. Right: mastoid-failure mode per subject—the pipeline falls back to common-average reference when a mastoid is bad and interpolates via spherical splines.

#### A.5 Behavioural compliance

Fig. 3 shows per-subject comprehension-question accuracy (mean 79 %; 13/16 above 70 %), the psychometric curve of accuracy vs. attended-speaker SNR, and the balance of attended-speaker labels across subjects. Two subjects fall below 65 %; per-trial correctness is released so downstream work can choose whether to filter low-compliance trials.

#### A.6 Cross-modal predictability extended table

Top-ranked gaze, IMU, video, and audio features by held-out Pearson  $r$  of a per-subject ridge regression  $X_{\text{EEG-spec}} \rightarrow y_{\text{feature}}$ , pooled over 16 subjects.

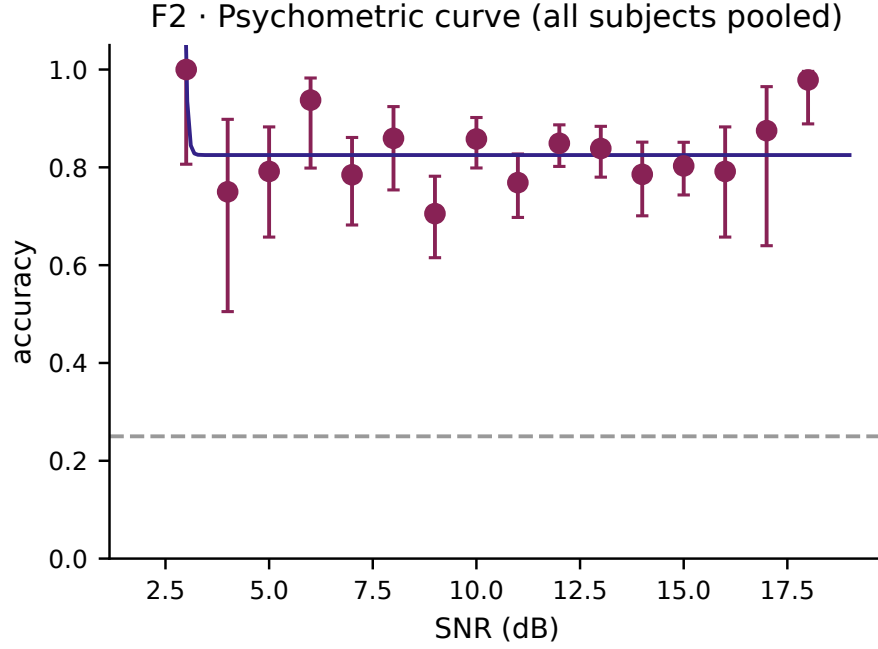


Figure 3: Behavioural compliance. Per-subject comprehension accuracy (left), psychometric accuracy-vs-SNR (middle), and attended-speaker label distribution (right). Per-trial correctness is released.

| Modality / feature           | Pooled $r$ | 95 % CI          |
|------------------------------|------------|------------------|
| <i>Gaze (23 available):</i>  |            |                  |
| sacc_rate                    | +0.549     | [+0.467, +0.630] |
| gx_valid                     | +0.540     | [+0.404, +0.672] |
| R_pup                        | +0.419     | [+0.323, +0.512] |
| <i>IMU (35 available):</i>   |            |                  |
| gyr_mag_mean                 | +0.559     | [+0.468, +0.647] |
| total_rotation               | +0.556     | [+0.465, +0.644] |
| <i>Video (15 available):</i> |            |                  |
| motion_energy_mean           | +0.487     | [+0.403, +0.565] |
| motion_energy_std            | +0.471     | [+0.389, +0.550] |
| <i>Audio (20 available):</i> |            |                  |
| una_env_der_pos              | -0.086     | [-0.139, -0.038] |
| att_zcr                      | +0.057     | [+0.002, +0.116] |

455

456 Directional flow features (which way the camera panned) and per-trial audio summaries are at zero  
457 predictability from EEG: EEG encodes *motion amplitude*, not motion direction, and the AAD signal  
458 is not a per-trial scalar audio summary.

Table 9: Per-subject hemisphere accuracy (T1) for every non-deep baseline. Env-CCA = CCA mel-28 30-s window. Spec-LR = 368-D spectral LogReg. Spec-GB = spectral LightGBM. Gaze = 23-D gaze-only LogReg. IMU = 35-D IMU LogReg. Video = 15-D scene-video LogReg. Fuse = calibrated late fusion of spectral-LR and gaze-LR. Chance = 0.5.

| Subject       | Env-CCA     | Spec-LR     | Spec-GB     | Gaze        | IMU         | Video       | Fuse        |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 1             | 0.55        | 0.48        | 0.40        | 0.41        | 0.56        | 0.44        | 0.44        |
| 2             | 0.60        | 0.90        | 0.75        | 0.93        | 0.73        | 0.81        | 0.97        |
| 3             | 0.45        | 0.94        | 0.91        | 1.00        | 0.82        | 0.71        | 0.98        |
| 4             | 0.51        | 0.88        | 0.81        | 0.98        | 0.81        | 0.82        | 0.94        |
| 5             | 0.53        | 0.70        | 0.60        | 0.74        | 0.64        | 0.68        | 0.70        |
| 6             | 0.54        | 0.73        | 0.62        | 0.65        | 0.62        | 0.58        | 0.71        |
| 7             | 0.59        | 0.89        | 0.86        | 0.92        | 0.61        | 0.70        | 0.98        |
| 8             | 0.52        | 0.62        | 0.63        | 0.57        | 0.43        | 0.61        | 0.57        |
| 9             | 0.54        | 0.60        | 0.60        | 0.67        | 0.47        | 0.46        | 0.59        |
| 10            | 0.51        | 0.66        | 0.60        | 0.87        | 0.62        | 0.49        | 0.78        |
| 11            | 0.57        | 0.43        | 0.51        | 0.69        | 0.33        | 0.45        | 0.46        |
| 12            | 0.54        | 0.60        | 0.50        | 0.59        | 0.54        | 0.57        | 0.57        |
| 13            | 0.51        | 0.77        | 0.65        | 0.85        | 0.68        | 0.68        | 0.86        |
| 14            | 0.47        | 0.61        | 0.55        | 0.67        | 0.48        | 0.59        | 0.62        |
| 15            | 0.46        | 0.92        | 0.82        | 0.98        | 0.71        | 0.74        | 0.97        |
| 16            | 0.47        | 0.72        | 0.67        | 0.84        | 0.65        | 0.55        | 0.78        |
| <b>Pooled</b> | <b>0.52</b> | <b>0.72</b> | <b>0.65</b> | <b>0.77</b> | <b>0.61</b> | <b>0.62</b> | <b>0.74</b> |

#### 459 A.7 Per-subject $\times$ baseline $\times$ task accuracy

#### 460 A.8 SNR-stratified accuracy

|                               | $\leq 6$ dB (n=112) | 7–10 (n=431) | 11–14 (n=736) | $\geq 15$ (n=320) |
|-------------------------------|---------------------|--------------|---------------|-------------------|
| <i>Envelope CCA mel-28:</i>   |                     |              |               |                   |
| T1 hemisphere                 | 0.464               | 0.487        | 0.499         | 0.525             |
| T2 inner-outer                | 0.545               | 0.524        | 0.503         | 0.538             |
| T3 4-class                    | 0.250               | 0.269        | 0.235         | 0.284             |
| <i>Spectral 368-D LogReg:</i> |                     |              |               |                   |
| T1 hemisphere                 | 0.528               | 0.670        | 0.636         | 0.548             |
| T2 inner-outer                | 0.598               | 0.556        | 0.564         | 0.533             |
| T3 4-class                    | 0.320               | 0.363        | 0.372         | 0.307             |
| <i>Gaze 23-D LogReg:</i>      |                     |              |               |                   |
| T1 hemisphere                 | 0.715               | 0.730        | 0.738         | 0.733             |
| T3 4-class                    | 0.544               | 0.512        | 0.473         | 0.442             |
| <i>EEG+gaze early fusion:</i> |                     |              |               |                   |
| T1 hemisphere                 | 0.652               | 0.723        | 0.731         | 0.674             |
| T3 4-class                    | 0.411               | 0.444        | 0.443         | 0.423             |

462 Source: results/generalisation\_snr.parquet and results/new\_experiments/C\_snr\_stratified.parquet.

#### 463 A.9 Window sweep (continuous AAD)

464 Backward-TRF CCA mel-28 accuracy by decision window (pooled across 16 subjects; each point is  
465 the mean over sub-windows within the 30-s trial). ITR follows Wolpaw assuming 2 classes.

| Window (s)                 | 1     | 2     | 4     | 8     | 16    | 30    |
|----------------------------|-------|-------|-------|-------|-------|-------|
| CCA mel-28 (pooled)        | 0.500 | 0.503 | 0.502 | 0.508 | 0.524 | 0.539 |
| ITR upper bound (bits/min) | 0.00  | 0.00  | 0.00  | 0.01  | 0.04  | 0.06  |

#### 467 A.10 Learning curves

468 Per-subject accuracy as a function of training-trial count  $n_{\text{train}} \in \{10, 20, 40, 60, 80\}$ , ten random  
469 resamples per  $n$ , within the 5-fold held-out structure.

470

|                   | 10    | 20    | 40    | 60    | 80    |
|-------------------|-------|-------|-------|-------|-------|
| Gaze (T1)         | 0.709 | 0.730 | 0.747 | 0.755 | 0.764 |
| Spectral (T1)     | 0.579 | 0.614 | 0.653 | 0.678 | 0.688 |
| Early fusion (T1) | 0.628 | 0.648 | 0.703 | 0.730 | 0.747 |
| Gaze (T3)         | 0.343 | 0.466 | 0.510 | 0.526 | 0.541 |
| Spectral (T3)     | 0.267 | 0.373 | 0.415 | 0.426 | 0.428 |
| Early fusion (T3) | 0.318 | 0.367 | 0.447 | 0.476 | 0.489 |

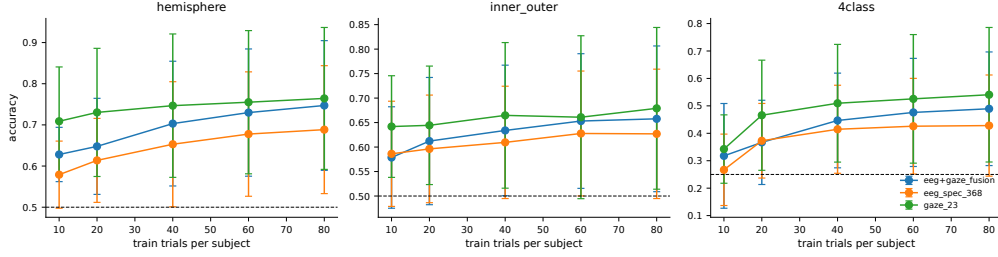


Figure 4: Learning curves per decoder per task. Error bars are per-subject std over 10 bootstrap resamples of the  $n_{\text{train}}$  subset.

#### 471 A.11 Partial motion-residualisation sweep

472  $X_\alpha = X - \alpha M(Z)$ . The curve is smooth and approximately linear on T1 (0.716 at  $\alpha=0$  to 0.521  
 473 at  $\alpha=1$ , no plateau)—every additional unit of motion-orthogonalisation continues to cost accuracy.  
 474 Consistent with genuinely entangled subspaces rather than a clean separable component. Source:  
 475 `results/new_experiments/D_partial_motion.parquet`.

476

| $\alpha$       | 0.00  | 0.25  | 0.50  | 0.75  | 1.00  |
|----------------|-------|-------|-------|-------|-------|
| T1 hemisphere  | 0.716 | 0.678 | 0.625 | 0.562 | 0.521 |
| T2 inner-outer | 0.650 | 0.627 | 0.579 | 0.538 | 0.510 |
| T3 4-class     | 0.442 | 0.412 | 0.359 | 0.312 | 0.270 |

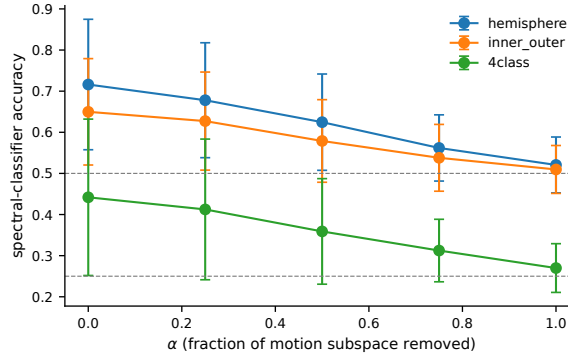


Figure 5: Partial motion-residualisation sweep. Error bars are std across 16 subjects. Slopes are monotone for all three tasks: no portion of the spectral-classifier signal is cleanly motion-orthogonal.

#### 477 A.12 Subject-quality composite and individual differences

478 A pre-registered quality composite (z-scored sum of comprehension accuracy, pupil-diameter mean,  
 479 gaze validity, alpha lateralisation, ISC at Cz) correlates with gaze hemisphere accuracy at  $r=+0.81$ ,  
 480 spectral-classifier hemisphere at  $r=+0.74$ , and calibrated-fusion hemisphere at  $r=+0.81$  ( $n=16$ ):  
 481 “good subject” is a single latent axis. The composite is a  $-0.41$  predictor of envelope-CCA hemi-  
 482 sphere: TRF accuracy rises when the subject’s behavioural/eye-tracker signal is *weaker*, consistent

with the behaviour- $\times$ -decoding finding (Sec. 6.4) that envelope tracking indexes effort rather than comprehension.

| Predictor        | Decoder       | Pearson $r$     |
|------------------|---------------|-----------------|
| pupil_mean       | gaze-T1       | +0.677          |
| comprehension    | gaze-T1       | +0.540          |
| gaze_valid       | gaze-T1       | +0.420          |
| head_motion      | EEG-T1 (TRF)  | +0.362          |
| pupil_mean       | EEG-T1 (spec) | -0.393          |
| bad_channel_rate | any decoder   | $ r  \leq 0.21$ |

A pre-registered high-quality tier (comprehension  $\geq 0.80$  AND Tobii gaze-validity  $\geq 0.70$ ) contains 8/16 subjects (S2, S3, S6, S7, S9, S13, S14, S15) and pools to T1 gaze 0.834, spectral 0.795, calibrated late fusion 0.835. Source: `results/new_experiments/E_quality_composite.parquet`, `results/evaluation_protocol/protocol_baseline_results.parquet`.

### A.13 Behaviour $\times$ decoding (extended)

Pooled  $2 \times 2$  contingency table ( $N = 1310$  behav-correct vs. 287 behav-wrong trials).

| Decoder                              | Task                | $\Delta$ pooled | $\chi^2$ | $p$   |
|--------------------------------------|---------------------|-----------------|----------|-------|
| Backward-TRF broadband               | envelope-attendance | -0.050          | 2.17     | 0.141 |
| Backward-TRF mel-28                  | envelope-attendance | +0.020          | 0.29     | 0.589 |
| Backward-TRF split $\delta + \theta$ | envelope-attendance | -0.060          | 3.15     | 0.076 |
| Backward-TRF CCA-mel                 | envelope-attendance | +0.000          | 0.00     | 1.000 |
| LOSO CCA mel-28                      | envelope-attendance | -0.027          | 0.57     | 0.451 |
| CCA 4-class (4-way)                  | T3                  | +0.005          | 0.01     | 0.923 |

The within-subject Wilcoxon test (Sec. 6.4) is the trustworthy one—pooled tables mix subject-level and trial-level variance, and subjects differ substantially in both comprehension rate (0.61–0.92) and decoder accuracy. Source: `results/behaviour_decoding*.parquet`, `results/new_experiments/F_behaviour_spectral.parquet`.

### A.14 LOSO transfer (extended)

| Model                              | T1 hemisphere | T2 inner-outer | T3 4-class |
|------------------------------------|---------------|----------------|------------|
| Envelope CCA mel-28 (backward-TRF) | 0.539         | —              | —          |
| Gaze 23-D LogReg                   | 0.715         | 0.607          | 0.449      |
| Spectral 368-D LogReg              | 0.615         | 0.572          | 0.352      |
| EEG+gaze concatenated LogReg       | 0.696         | 0.614          | 0.435      |

Gaze dominates cross-subject; the spectral classifier still gives +7.6pp over envelope CCA on T1, showing the spectral feature set does not over-fit individual EEG topographies as much as a pure TRF does. EEG+gaze concatenation under-performs gaze-only because it adds a weaker modality; a calibrated late-fusion scheme tracks the stronger modality (Sec. 5.4). Source: `results/loso_cca_mel28/s*.parquet`, `results/new_experiments/A_loso_spectral.parquet`, `results/evaluation_protocol/protocol_baseline_res`

### A.15 Neuroscience validation

The dataset reproduces canonical AAD neuroscience markers: (a) FOOOF aperiodic slope at mid-parietal electrodes is  $\approx 1.4$  (s.d.  $< 0.15$  across subjects), within the adult-awake range; (b) theta-gamma phase-amplitude coupling peaks at  $\theta \approx 6.5$  Hz /  $\gamma \approx 35$  Hz (canonical); (c) microstate k-means ( $k = 4$ ) recovers the canonical A/B/C/D morphologies with isotropic coverage; (d) inter-subject correlation at Cz averages  $r = 0.023$  but peaks at  $r = 0.079$  on prosody climaxes (trial 7). Alpha lateralisation across the cohort is weak ( $t = 0.29$ ,  $p = 0.77$ ,  $n = 16$ , 10-trial subset)—consistent with recent findings that the canonical alpha-lat AAD signature is paradigm-sensitive.



## 513 A.16 Reproducing every number in this paper

514 Every headline number is produced by one of the scripts below. All scripts read the  
515 raw data at /fs/ess/PAS2301/Data/AAD Data Collection/ and write parquets under  
516 analysis/results/. The SLURM account is configurable via -account.

```
517 # Preliminaries (one-off cache build; ~1 h on 16-core node).
518 cd analysis
519 sbatch slurm_scripts/iter5_verify.sbatch      # 368-D spectral features + logreg
520 sbatch slurm_scripts/iter6.sbatch EXP=imu     # 35-D IMU features + logreg
521 sbatch slurm_scripts/iter6.sbatch EXP=video   # 15-D video features + logreg
522 sbatch slurm_scripts/iter6.sbatch EXP=clean_all # EEG-cleaning ablation
523 sbatch slurm_scripts/aad_v3.sbatch FEATURES=cca_mel # envelope CCA 30-s
524 sbatch slurm_scripts/aad_v3.sbatch FEATURES=broadband
525 sbatch slurm_scripts/aad_v3.sbatch FEATURES=mel28
526 sbatch slurm_scripts/aad_v3.sbatch FEATURES=split_delta_theta
527
528 # Controls and sweeps.
529 sbatch slurm_scripts/iter4.sbatch EXP=gaze_residualised
530 sbatch slurm_scripts/iter8.sbatch              # trial-level motion residualisation
531 sbatch slurm_scripts/aad_loso.sbatch           # envelope LOSO
532 sbatch slurm_scripts/new_experiments.sbatch    # (A)-(G), see below
533
534 # Aggregate and produce paper numbers.
535 python scripts/aggregate_iter5.py
536 python scripts/aggregate_iter4.py
537 python scripts/aggregate_benchmark_leaderboard.py
538 python scripts/evaluation_protocol.py
539
540 # Figures.
541 jupyter execute 10_publication_figures.ipynb
```

542 Per-experiment mapping to tables/figures in this paper:

| 543 | Table/Fig.                   | Produced by                                                    | Output parquet                                                   |
|-----|------------------------------|----------------------------------------------------------------|------------------------------------------------------------------|
|     | Table 2 (TRF)                | aggregate_benchmark_leaderboard.py                             | benchmark_leaderboard_pooled.parquet                             |
|     | Table 3 (spectral)           | aggregate_iter5.py                                             | iter5_summary.parquet                                            |
|     | Table 4 (fusion)             | aggregate_iter5.py +<br>new_experiments.py::G                  | iter5_fusion.parquet +<br>G_calibrated_late_fusion.parquet       |
|     | Table 5 (cleaning)           | aggregate_iter6.py                                             | iter6_cleaning.parquet                                           |
|     | Table 6 (cross-modal)        | aggregate_iter7.py                                             | iter7_cross_modal.parquet                                        |
|     | Table 7 (residualisation)    | aad_artefact_subtracted.py                                     | iter8/s*.parquet                                                 |
|     | Table 8 (behav ×)            | aggregate_behaviour_decoding.py                                | behaviour_decoding_within_subject.parquet                        |
|     | Fig. 2                       | plot_bad_channels.py                                           | bad_channels*.parquet                                            |
|     | Fig. 3                       | notebook 02                                                    | 02_per_subject_acc.parquet                                       |
| 544 | Fig. 1                       | aggregate_benchmark_leaderboard.py                             | benchmark_leaderboard.parquet                                    |
|     | App. A.8 SNR                 | aggregate_generalisation.py                                    | generalisation_snr.parquet                                       |
|     | App. A.9 windows             | aggregate_continuous_aad.py                                    | continuous_aad_window_sweep.parquet                              |
|     | App. A.10 learning curve     | new_experiments.py::B                                          | B_learning_curve.parquet                                         |
|     | App. A.11 partial            | new_experiments.py::D                                          | D_partial_motion.parquet                                         |
|     | App. A.12 quality            | aggregate_individual_differences.py +<br>new_experiments.py::E | individual_differences*.parquet +<br>E_quality_composite.parquet |
|     | App. A.13 behaviour (pooled) | aggregate_behaviour_decoding.py +<br>new_experiments.py::F     | behaviour_decoding*.parquet +<br>F_behaviour_spectral.parquet    |
|     | App. A.14 LOSO               | aad_loso.py +<br>new_experiments.py::A                         | loso_cca_mel28/s*.parquet +<br>A_loso_spectral.parquet           |
|     | Sec. 4 protocol              | evaluation_protocol.py                                         | evaluation_protocol/                                             |

545 **A.17 Evaluation-harness minimal example**

```
546 from scripts.evaluation_protocol import evaluate, baseline_logreg, ProtocolResult
547 import pandas as pd
548 features_df = pd.read_parquet("results/eeg_spectral/s1.features.parquet") # ... subjects 1..16
549 def features_fn(s):
550     g = full_df[full_df.subject == s]
551     return g[feat_cols].values, g.attended.values, g.trial.values
552 results = evaluate(my_predictor_fn, features_fn, full_df, subset_name="all")
553 # --> list[ProtocolResult], one per (task, split, window, subset). Dump to parquet
554 #     and upload alongside your paper.
```

555 The evaluate() function returns one row per (task, split) combination with mean\_acc, ci\_lo,  
556 ci\_hi, n\_subjects, chance, so a new paper's reporting is forced to fit the same schema.

## NeurIPS Paper Checklist

### 1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Sec. 1 list the five concrete contributions (multimodal corpus, reference pipeline, benchmark suite, motion-residualisation control, Croissant metadata); each is supported by a numbered section in the paper.

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Sec. 9 discusses cohort size, paradigm specificity, the S1 hardware outlier, the absence of deep-model ceilings in the submitted numbers, and the shared-variance confound exposed by the motion-residualisation control.

### 3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper presents a dataset and empirical benchmarks; there are no theoretical results requiring proof.

### 4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Sec. 3–6 specify the preprocessing (bandpass, notch, auto-reference, spherical-spline interpolation), feature definitions (368-dim spectral vector, 23-dim gaze vector, 35-dim IMU vector, 15-dim video vector), classifier choices (L2-LogReg, LightGBM  $n=300$ ), CV protocol (5-fold stratified within-subject, bootstrap 95 % CI over the 16 per-subject accuracies), and the residualisation procedure.

### 5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Sec. 7 specifies anonymous hosting URLs for data and code at submission, a CC-BY-NC-4.0 licence on the recordings with CC-0 on the LibriVox stimuli, and a Croissant RecordSet with RAI fields. **[TODO]**Final hosting location (Hugging Face/OpenNeuro/Zenodo) and persistent DOI to be set before camera-ready.

### 6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Sec. 5.1–5.4 specify the ridge  $\alpha$  grid ( $10^{-1} \dots 10^6$ , 12 log-spaced points, inner 5-fold CV on  $\rho_{\text{att}} - \rho_{\text{una}}$ ), TRF lag window ( $-200$  to  $+500$  ms at 25 ms steps), LightGBM hyperparameters, 5-fold stratified CV per subject with bootstrap 95 % CI over 16 subjects, and decision-window sweep  $\{1, 2, 4, 8, 16, 30\}$  s.

### 7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

607 Answer: [Yes]  
 608 Justification: All pooled accuracies are accompanied by 95 % bootstrap CIs (resampling the  
 609 16 per-subject accuracies with replacement). Sec. 6.3 reports paired Wilcoxon signed-rank  
 610  $p$ -values against the raw condition.

611 **8. Experiments compute resources**

612 Question: For each experiment, does the paper provide sufficient information on the com-  
 613 puter resources (type of compute workers, memory, time of execution) needed to reproduce  
 614 the experiments?

615 Answer: [Yes]

616 Justification: All submitted baselines run CPU-only on an OSC SLURM cpu partition, one  
 617 task per (subject×backbone), 8 CPUs and 32 GB memory per task, 2 h wall-time limit. A  
 618 full benchmark sweep of 16 subjects × 4 TRF backbones + spectral + gaze + IMU + video  
 619 fits in ~20 minutes wall-clock when parallelised across nodes. Deep-model baselines listed  
 620 in Sec. 9 are GPU-bound and not included in the submitted numbers.

621 **9. Code of ethics**

622 Question: Does the research conducted in the paper conform, in every respect, with the  
 623 NeurIPS Code of Ethics?

624 Answer: [Yes]

625 Justification: Sec. 8 describes the IRB approval, informed consent specifically covering  
 626 multimodal data release, face-blurring of bystanders in egocentric scene video, compensation  
 627 at jurisdictional minimum wage, and access-gated raw video.

628 **10. Broader impacts**

629 Question: Does the paper discuss both potential positive societal impacts and negative  
 630 societal impacts of the work performed?

631 Answer: [Yes]

632 Justification: Sec. 8 discusses the intended research use (hearing-aid algorithms, BCI re-  
 633 search, audio-cortex neuroscience), the foreseeable misuses (trait inference from behavioural  
 634 data; AAD decoders in surveillance contexts), and mitigations (CC-BY-NC-4.0 licence,  
 635 access-by-request raw video, face-blurring).

636 **11. Safeguards**

637 Question: Does the paper describe safeguards that have been put in place for responsible  
 638 release of data or models that have a high risk for misuse (e.g., pre-trained language models,  
 639 image generators, or scraped datasets)?

640 Answer: [Yes]

641 Justification: Sec. 8: face-blurring of incidental persons in egocentric video, CC-BY-NC-4.0  
 642 licence restricting commercial reuse, and access-by-request gating of the raw scene-camera  
 643 footage. No pre-trained generative models are released.

644 **12. Licenses for existing assets**

645 Question: Are the creators or original owners of assets (e.g., code, data, models), used in  
 646 the paper, properly credited and are the license and terms of use explicitly mentioned and  
 647 properly respected?

648 Answer: [Yes]

649 Justification: Sec. 7 credits the LibriVox public-domain corpus (CC-0) for audio stimuli  
 650 and lists the MNE-Python, NumPy, scikit-learn, LightGBM, and PyTorch versions used  
 651 in the pipeline (complete list in the environment file). **[TODO]**Add explicit CTAN / PyPI  
 652 references in the camera-ready acknowledgements.

653 **13. New assets**

654 Question: Are new assets introduced in the paper well documented and is the documentation  
 655 provided alongside the assets?

656 Answer: [Yes]

657 Justification: The dataset is packaged with a Croissant RecordSet, a per-trial provenance  
658 manifest (`results/*.parquet`), loader code with Jupyter notebook demos for every task  
659 in Sec. 5, and a README describing the directory layout and trial-index conventions  
660 (App. A.2).

661 **14. Crowdsourcing and research with human subjects**

662 Question: For crowdsourcing experiments and research with human subjects, does the paper  
663 include the full text of instructions given to participants and screenshots, if applicable, as  
664 well as details about compensation (if any)?

665 Answer: [Yes]

666 Justification: Sec. 3 describes the paradigm (cued attention across four spatial talkers, 30-s  
667 trials, per-trial comprehension question). Sec. 8 states compensation was at the jurisdictional  
668 minimum wage. [TODO]Supplementary material will include the verbatim participant-  
669 instruction script and a screenshot of the comprehension-question interface.

670 **15. Institutional review board (IRB) approvals or equivalent for research with human**  
671 **subjects**

672 Question: Does the paper describe potential risks incurred by study participants, whether  
673 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)  
674 approvals (or an equivalent approval/review based on the requirements of your country or  
675 institution) were obtained?

676 Answer: [Yes]

677 Justification: The study was IRB-approved and informed consent was obtained specifically  
678 covering the release of de-identified EEG, eye-tracking, IMU, and head-mounted scene  
679 video. Institution and protocol number are withheld during anonymous review and will be  
680 added in the camera-ready version (Sec. 8).

681 **16. Declaration of LLM usage**

682 Question: Does the paper describe the usage of LLMs if it is an important, original, or  
683 non-standard component of the core methods in this research?

684 Answer: [N/A]

685 Justification: LLMs play no role in data collection, preprocessing, feature extraction, baseline  
686 classifiers, or the motion-residualisation control. Writing and formatting assistance are not  
687 required to be declared per the NeurIPS policy.